This assignment will not be graded and is only for practice.

Level 1

1) Consider two vectors $a = (2, 0, 3, 0, 8)$, $b = (3, 2, -2, 4, 0)$ in $\mathbb{R}^5$. Choose the set of correct options. ***1 point***

- [ ] a and b are orthogonal.
- [ ] a and b are not orthogonal.
- [ ] $(a - b) \cdot a = 0$.
- [ ] $(a - b) \cdot a = 77$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

a and b are orthogonal.
$(a - b) \cdot a = 77$.

2) Choose the set of correct statements. ***1 point***

- [ ] In an orthogonal set, the norms of all the vectors are equal.
- [ ] In an orthogonal set, the vectors are linearly independent.
- [ ] In an orthogonal set, the vectors are linearly dependent.
- [ ] If the columns of an $n \times n$ coefficient matrix A comprises of the individual vectors of an orthogonal set in $\mathbb{R}^n$, then there must be a unique solution to the system $AX = b$, where X, b are $n \times 1$ vectors.
- [ ] If the columns of an $n \times n$ coefficient matrix A comprises of the individual vectors of an orthogonal set in $\mathbb{R}^n$, then there are no solutions to the system $AX = b$, where X, b are $n \times 1$ vectors.
- [ ] The determinant of a square matrix formed by a set of orthogonal vectors in $\mathbb{R}^n$ is zero.
- [ ] A set of n vectors can never form an orthogonal basis in $\mathbb{R}^{n-1}$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

In an orthogonal set, the vectors are linearly independent.
If the columns of an $n \times n$ coefficient matrix A comprises of the individual vectors of an orthogonal set in R^n, then there must be a unique solution to the system $AX = b$, where X, b are $n \times 1$ vectors.
A set of n vectors can never form an orthogonal basis in R^{n-1}.

3) Which of the following is an orthogonal basis of the given vector spaces with respect to the standard inner product (dot product)? ***1 point***

- [ ] $\{(1, 0), (0, 1)\}$ is an orthogonal basis of R^2.
- [ ] $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is an orthogonal basis of R^3.
- [ ] $\{(3, 4), (4, -3), (2, -3)\}$ is an orthogonal basis of R^2.
- [ ] $\{(2, 1, -1), (-1, 1, -1), (3, -3, 3)\}$ is an orthogonal basis of R^3.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\{(1, 0), (0, 1)\}$ is an orthogonal basis of R^2.
$\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is an orthogonal basis of R^3.

Level 2

4) Find a vector in R^4 that is orthogonal to the subspace spanned by $(1, 1, 0, 0)$ and $(0, 1, 1, 0)$ with respect to the dot product as the inner product. ***1 point***

- [ ] (1,-1,1,0)
- [ ] (2,3,4,5)
- [ ] (1,-1,1,1)

- [ ] (1,1,1,0)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(1,-1,1,0)
(1,-1,1,1)

5) Which of the following are orthogonal to the vector $(1, 2)$ in $\mathbb{R}^2$ with respect to the inner product $< v, w >= v_1w_1 - v_1w_2 - v_2w_1 + 4v_2w_2$? ***1 point***

- [ ] $(-7, 1)$
- [ ] $(6, -1)$
- [ ] $(7, 1)$
- [ ] $(-9, 1)$
- [ ] $(14, 2)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(7, 1)$
$(14, 2)$

6) Consider the system of linear equations:

$x_1 - 2x_2 + 3x_3 = 1$

$2x_1 + x_2 = 5$

$3x_1 - 6x_2 - 5x_3 = 9.$

Number of solutions of the above system is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

7) Let A be the coefficient matrix of the system of linear equations: ***1 point***

$x_1 - 2x_2 + 3x_3 = 1$

$2x_1 + x_2 = 5$

$3x_1 - 6x_2 - 5x_3 = 9.$

Which one of the following is true about the matrix AA^T?

- [ ] A scalar matrix.
- [ ] The identity matrix.
- [ ] A diagonal matrix.
- [ ] A lower triangular matrix.
- [ ] An upper triangular matrix.
- [ ] None of the above.

No, the answer is incorrect.
Score: 0

Accepted Answers:

A diagonal matrix.
A lower triangular matrix.
An upper triangular matrix.

Your score is: 0/7